

Q1

GET /api/q1/welcome

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
'https://localhost:44378/api/q1/welcome' \
-H 'accept: */*'
```

Request URL

```
https://localhost:44378/api/q1/welcome
```

Server response

Code	Details
200	Response body <pre>Welcome to 5125!</pre> Response headers <pre>content-encoding: gzip content-type: text/plain; charset=utf-8 date: Wed, 05 Feb 2025 04:02:46 GMT server: Microsoft-IIS/10.0 vary: accept-encoding x-powered-by: ASP.NET</pre>

Responses

Code	Description	Links
200	OK	No links

```
C:\Users\aunku>curl -X GET "https://localhost:44378/api/q1/welcome"
Welcome to 5125!
C:\Users\aunku>
```

WebApplication1

Debug Any CPU IIS Express

Q1Controller.cs

```
namespace Assignment.Controllers
{
    [ApiController]
    [Route("api/q1")]
    public class Q1Controller : ControllerBase
    {
        /// <summary>
        /// Returns a welcome message.
        /// </summary>
        /// <returns>A welcome string message</returns>
        [HttpGet("welcome")]
        public IActionResult GetWelcome()
        {
            return Ok("Welcome to 5125!");
        }
    }
}
```

Solution Explorer

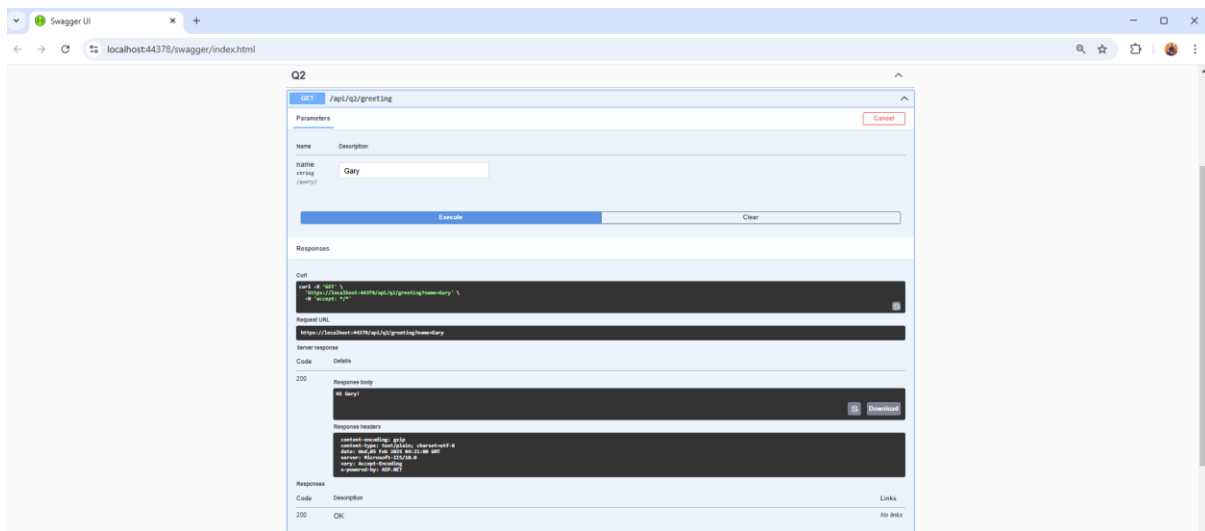
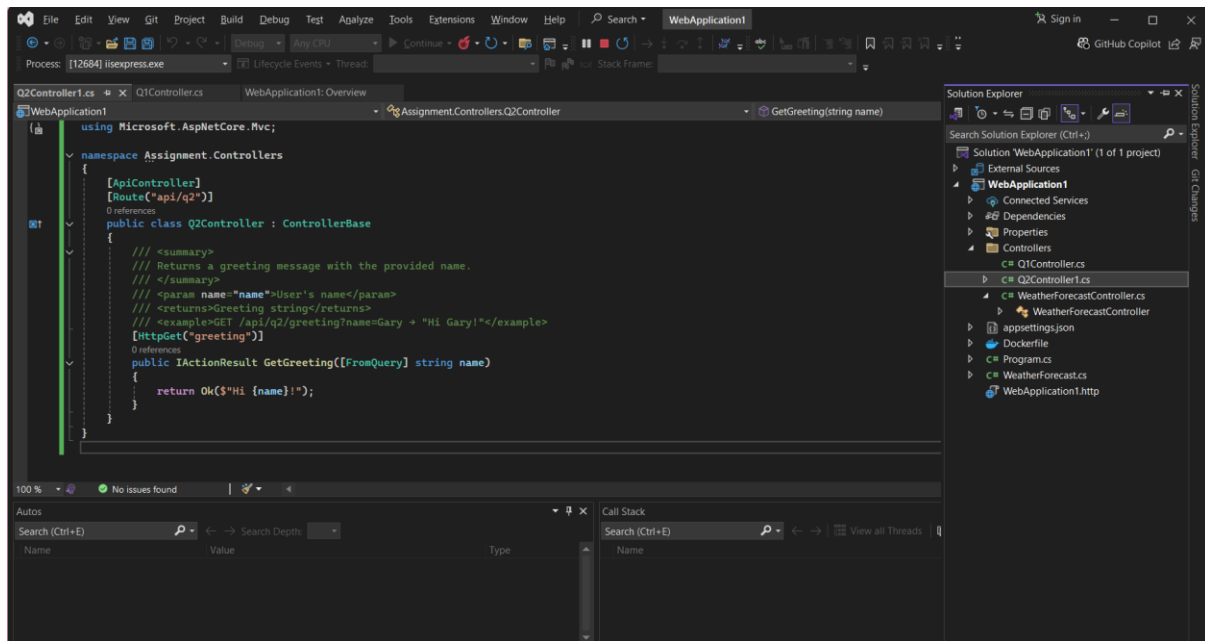
WebApplication1

- Connected Services
- Dependencies
- Properties
- Controllers
 - Q1Controller.cs
 - Q2Controller1.cs
 - Q3Controller.cs
 - Q4Controller.cs
 - Q5Controller.cs
 - Q6Controller.cs
 - Q7Controller.cs
 - Q8Controller.cs
 - WeatherForecastController.cs
- appsettings.json
- Dockerfile
- Program.cs
- WeatherForecast.cs
- WebApplication1.http

Output

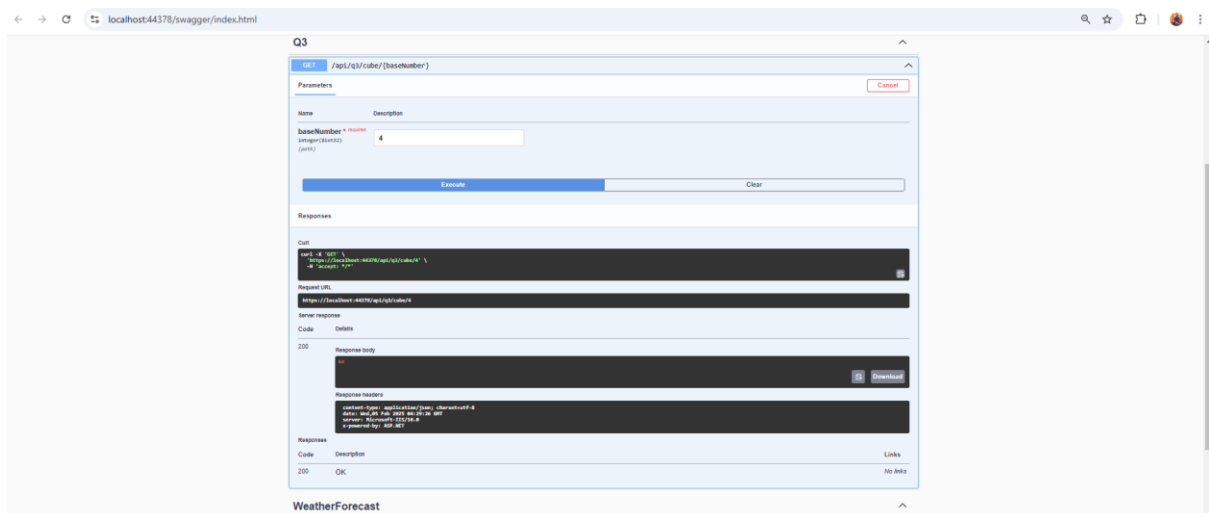
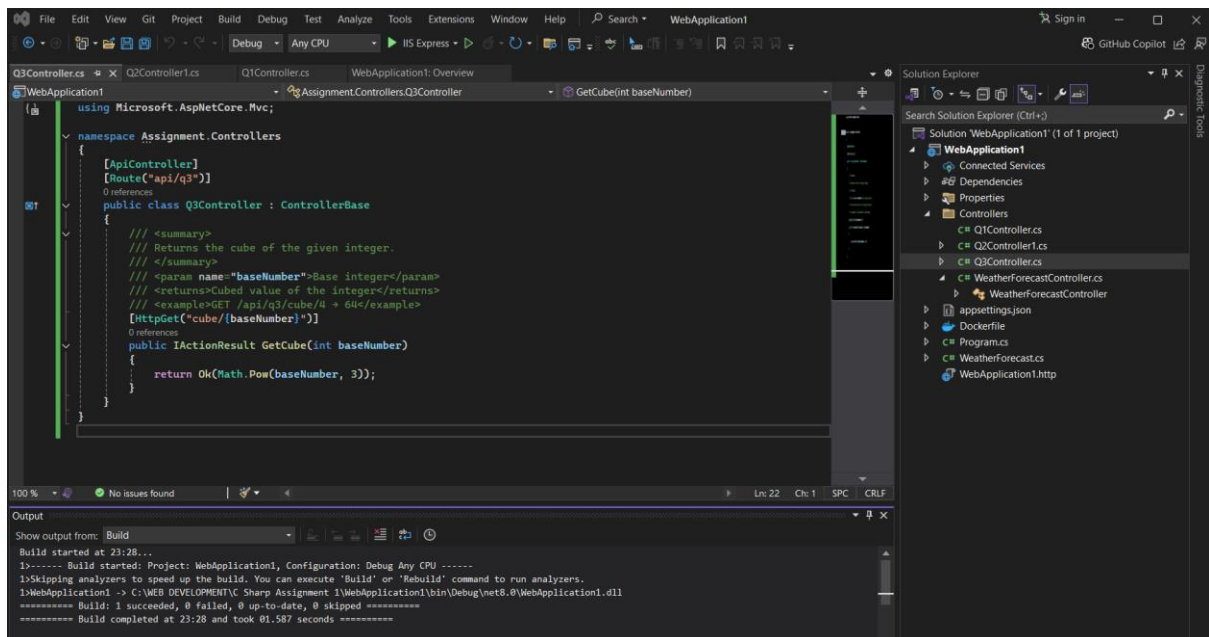
```
Build started at 23:55...
1>----- Build started: Project: WebApplication1, Configuration: Debug Any CPU -----
===== Build: 1 succeeded, 0 failed, 0 up-to-date, 0 skipped =====
===== Build completed at 23:55 and took 00.395 seconds =====
```

Q2)



```
C:\Users\aunku>curl -k "https://localhost:44378/api/q2/greeting?name=Gary"
Hi Gary!
C:\Users\aunku>
```

Q3)



```
C:\Users\auku>curl -k "https://localhost:44378/api/q3/cube/4"
64
C:\Users\auku>
```

Q4)

The screenshot shows the Visual Studio IDE with the `Q4Controller.cs` file open. The code is as follows:

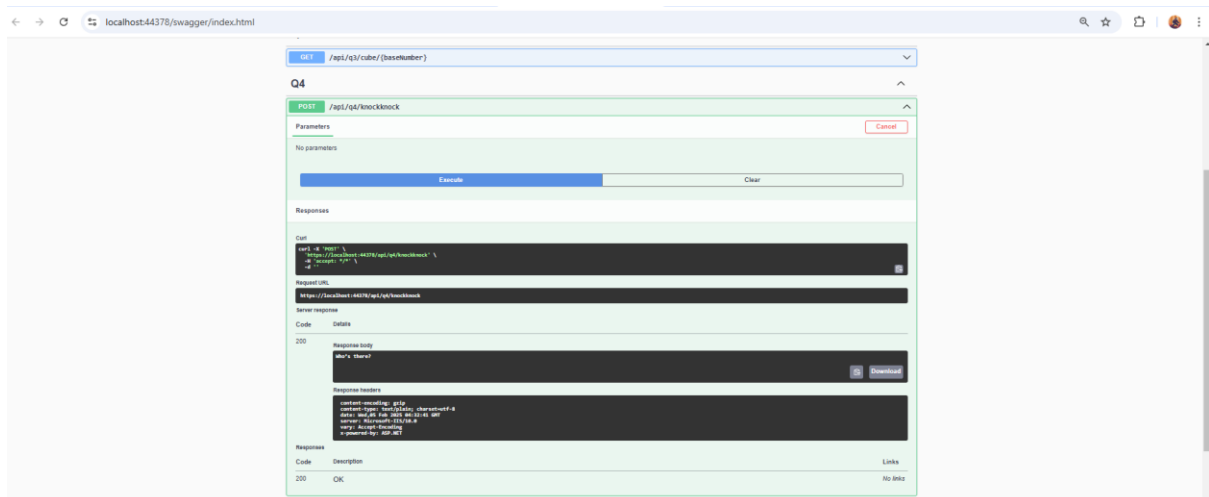
```
using Microsoft.AspNetCore.Mvc;

namespace Assignment.Controllers
{
    [ApiController]
    [Route("api/q4")]
    public class Q4Controller : ControllerBase
    {
        /// <summary>
        /// Returns the start of a knock-knock joke.
        /// </summary>
        /// <returns>"Who's there?"</returns>
        /// <example>POST /api/q4/knockknock - "Who's there?"</example>
        [HttpPost("knockknock")]
        public IActionResult PostKnockKnock()
        {
            return Ok("Who's there?");
        }
    }
}
```

The Solution Explorer on the right shows the project structure for `WebApplication1`, including controllers like `Q1Controller.cs`, `Q2Controller.cs`, `Q3Controller.cs`, `Q4Controller.cs`, and `WeatherForecastController.cs`.

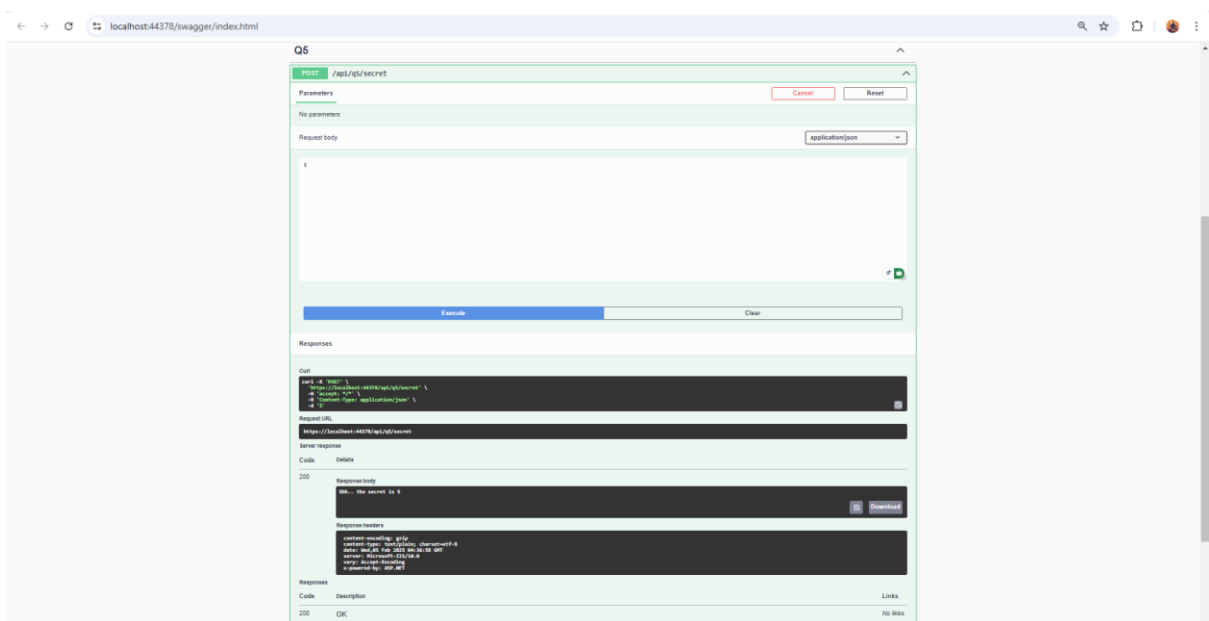
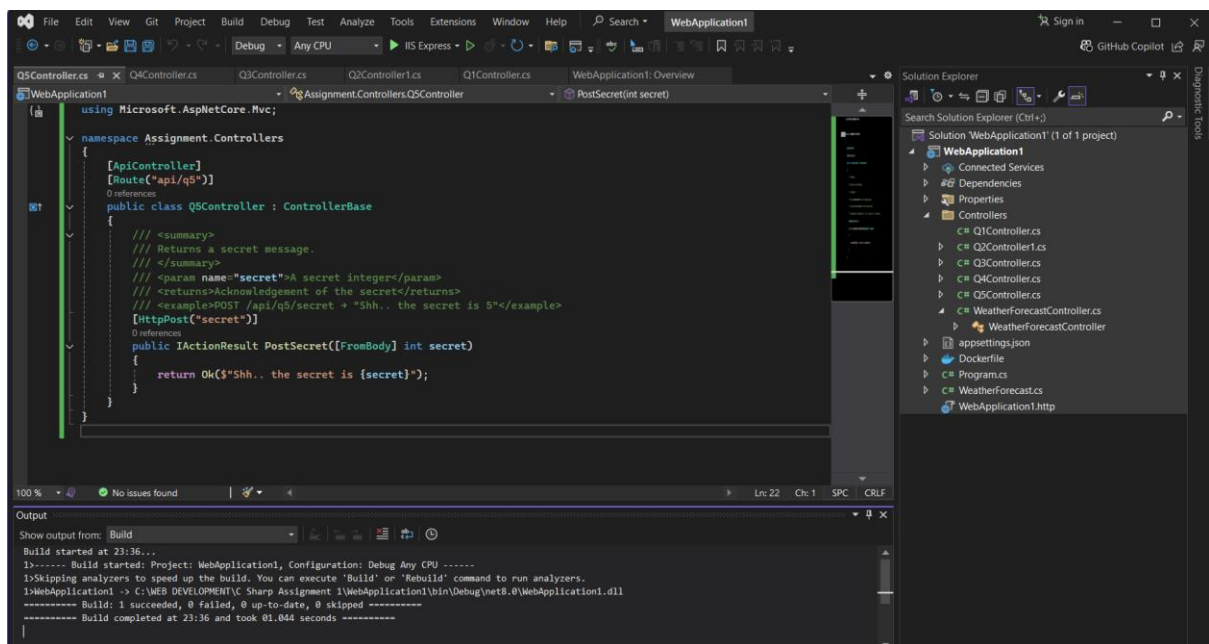
The Output window at the bottom shows the build process:

```
Build started at 23:31...
1>----- Build started: Project: WebApplication1, Configuration: Debug Any CPU -----
1>Skipping analyzers to speed up the build. You can execute 'Build' or 'Rebuild' command to run analyzers.
1>WebApplication1 -> C:\WEB DEVELOPMENT\Sharp Assignment 1\WebApplication1\bin\Debug\net8.0\WebApplication1.dll
----- Build: 1 succeeded, 0 failed, 0 up-to-date, 0 skipped -----
----- Build completed at 23:31 and took 01.013 seconds -----
```



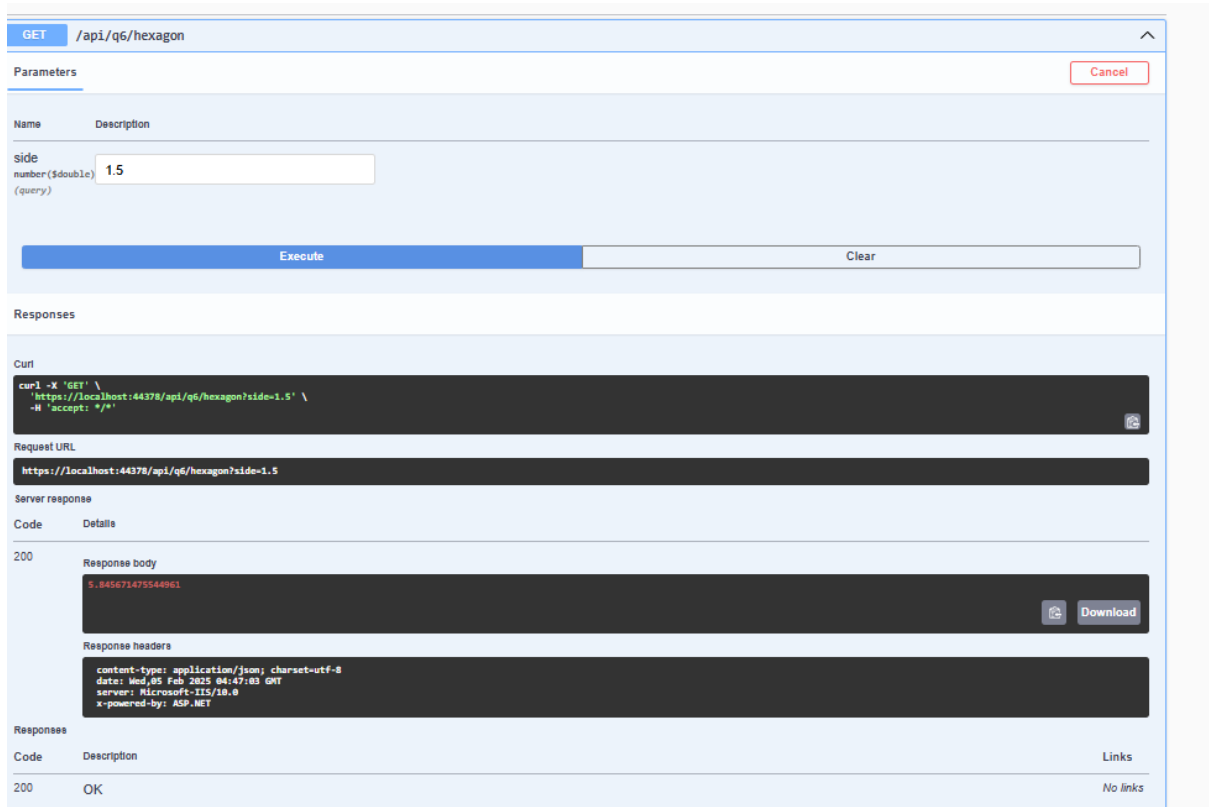
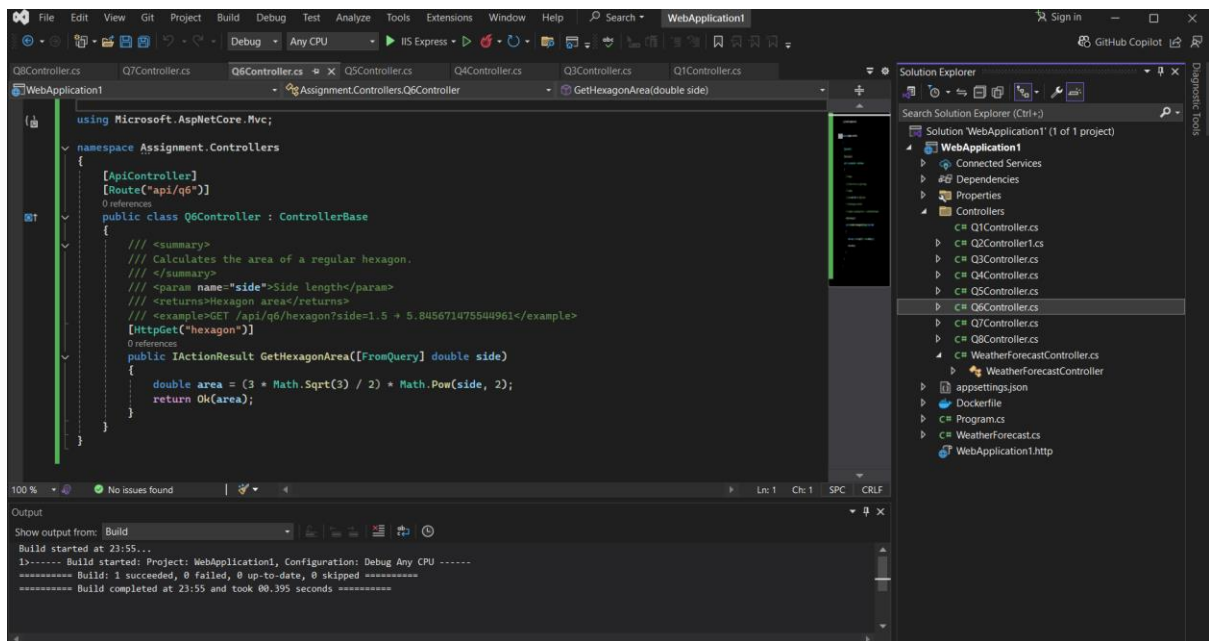
```
C:\Users\anunku>curl -k -X POST -H "Content-Length: 0" "https://localhost:44378/api/q4/knockknock"
Who's there?
C:\Users\anunku>
```

Q5)



```
C:\Users\anku>curl -k -H "Content-Type: application/json" -d "5" "https://localhost:44378/api/q5/secret"
Shh.. the secret is 5
C:\Users\anku>
```

Q6)



```
C:\Users\anunku>curl -k "https://localhost:44378/api/q6/hexagon?side=1.5"
5.845671475544961
C:\Users\anunku>
```

Q7)

The screenshot shows the Visual Studio IDE with the Q7Controller.cs file open. The code defines a controller that adjusts the current date by a given number of days. The Solution Explorer on the right shows the project structure, including the Controllers folder and the Q7Controller.cs file. The Output window at the bottom shows the build process, indicating that the build was successful.

```
using Microsoft.AspNetCore.Mvc;

namespace Assignment.Controllers
{
    [ApiController]
    [Route("api/q7")]
    public class Q7Controller : ControllerBase
    {
        /// <summary>
        /// Adjusts the current date by a given number of days.
        /// </summary>
        /// <param name="days">Number of days to adjust</param>
        /// <returns>Adjusted date as string (yyyy-MM-dd)</returns>
        /// <example>GET /api/q7/timemachine?days=5 + "2025-02-10"</example>
        [HttpGet("timemachine")]
        public IActionResult GetTimeMachine([FromQuery] int days)
        {
            return Ok(DateTime.Today.AddDays(days).ToString("yyyy-MM-dd"));
        }
    }
}
```

The screenshot shows the Swagger UI for the GET /api/q7/timemachine endpoint. The Parameters section shows a query parameter named 'days' of type 'integer (sint32)' with a value of 5. The Responses section shows a 200 status code with a response body of '2025-02-09'. The Request URL is 'https://localhost:44378/api/q7/timemachine?days=5'. The Server response section shows the response headers, including 'content-encoding: gzip', 'content-type: text/plain; charset=utf-8', 'date: Wed, 05 Feb 2025 04:51:28 GMT', 'server: Microsoft-IIS/10.0', 'vary: Accept-Encoding', and 'x-powered-by: ASP.NET'.

GET /api/q7/timemachine

Parameters

Name	Description
days	integer (sint32) (query)

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'https://localhost:44378/api/q7/timemachine?days=5' \
  -H 'accept: */*'
```

Request URL

https://localhost:44378/api/q7/timemachine?days=5

Server response

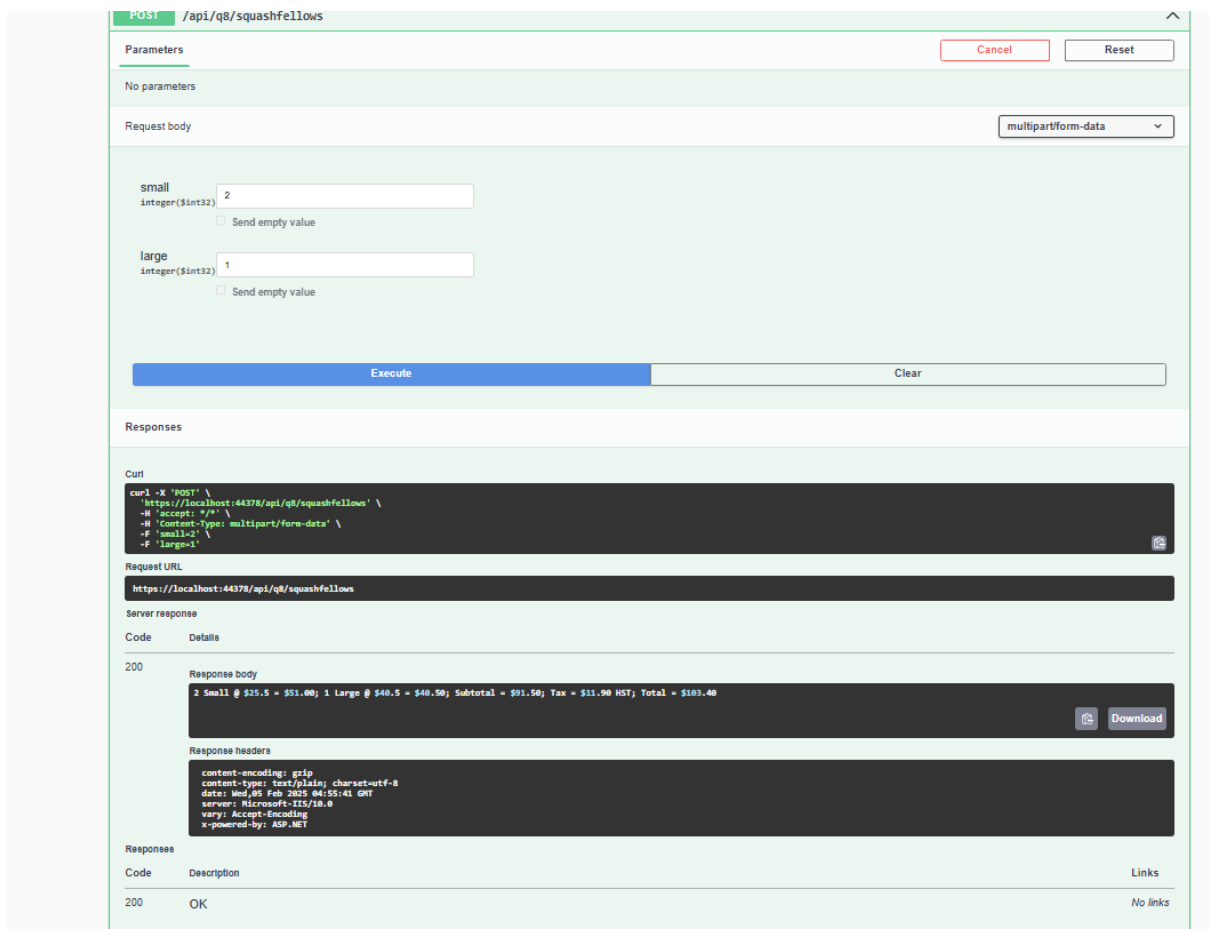
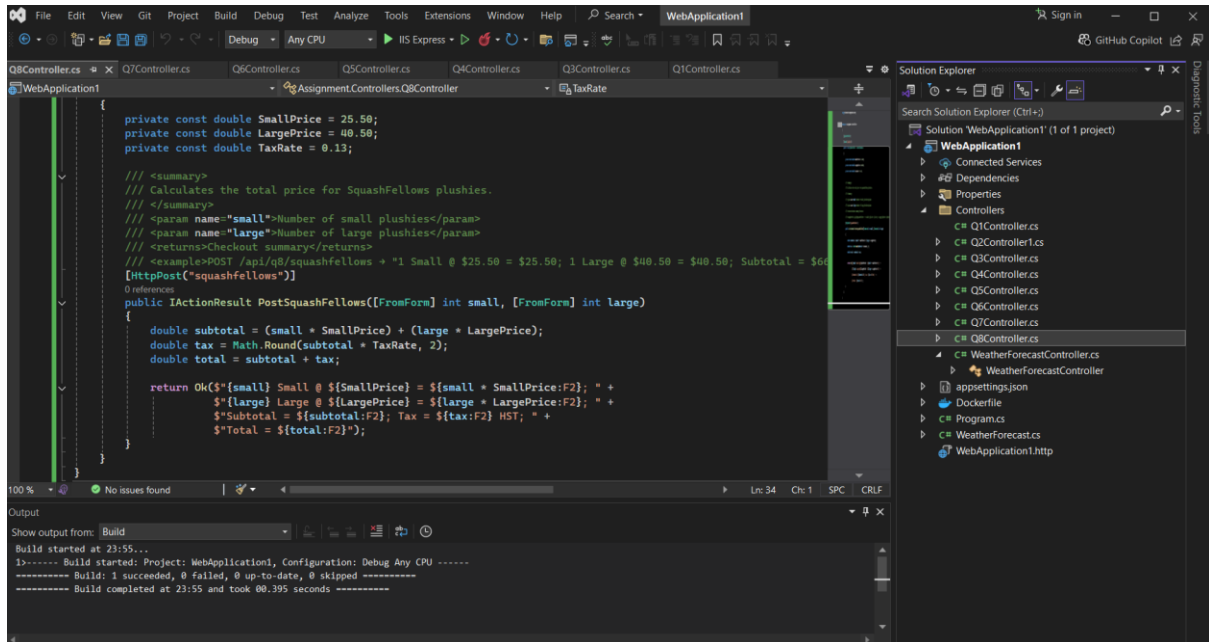
Code	Details
200	Response body 2025-02-09 Response headers content-encoding: gzip content-type: text/plain; charset=utf-8 date: Wed, 05 Feb 2025 04:51:28 GMT server: Microsoft-IIS/10.0 vary: Accept-Encoding x-powered-by: ASP.NET

Responses

Code	Description	Links
200	OK	No links

```
C:\Users\auunku>curl -k "https://localhost:44378/api/q7/timemachine?days=5"
2025-02-09
C:\Users\auunku>
```

Q8)




```
C:\Users\auunku>curl -k -X POST -d "Small=2&Large=1" "https://localhost:44378/api/q8/squashfellows"
2 Small @ $25.5 = $51.00; 1 Large @ $40.5 = $40.50; Subtotal = $91.50; Tax = $11.90 HST; Total = $103.40
C:\Users\auunku>
```